Caution !

Do not use an abrasive cleaner or allow solvents near the windows.

Do not use glass cleaning liquids, as they generally contain solvents and weak acids

Cleaning Source Viewing Window

In front of the viewing window, is a piece of toughened glass, which will become coated with aluminium over a period of time.

The light coating can be removed by using a razor blade type scraper. An alternative is to use a lint free cloth damped with a weak caustic acid solution.

Figure 8.1-21 Example of a technician accessing the glass window with the cover removed.

Caution !

Do not use wire wool, sand paper or abrasive cleaners, as the window will become scratched and difficult to view through.

Cleaning the Cryogenic Coils

Due to the fact that the cryogenic coils collect water on them, they also sometimes collect aluminium dust.

The coils must be wiped clean of this aluminium debris. If not the collection, efficiency of the coils will decrease and lead to longer pump down periods. A lint free cloth can be used to carry out this operation.

Caution !

Do not use any abrasive material to "polish" the coils as excess use could lead to reduced wall thickness and potential refrigerant leaks.

Cleaning Planar Plasma Treaters

The electrode plates should be inspected and cleaned as necessary. Film debris should be removed after any web break situation.

The cleaning interval is dependent upon the types of substrates being pre-treated. Many substrates have coatings, which can deposit on the electrode plates and plasma surfaces during treatment and require cleaning prior to running the unit again.

Aligmers coming off the substrate are particularly problematic and can initiate arcing events on the plasma treater electrode plates. As a general rule, if unacceptable levels of arcs occur, the cleanliness of the plates should be checked. Open the treater enclosure doors to inspect the electrodes and insulators.

Figure 8.1-22 Example of a plasma treater that has not been cleaned, showing erosion and allowing arcing to take place

Depending on the deposit or level of contamination on the plates, the level of cleaning required can differ dramatically. Use the least abrasive material possible to remove the source of contamination responsible for causing the arc events.

	**Note !**
	To prevent contamination, the use of protective gloves is essential throughout any contact with the plasma treater, particularly the electrode surfaces.

Basic Clean

Definition: A basic clean refers to a clean-up procedure, which removes any oxide build-up, general contamination (dust/dirt) and/or arc marks on any visible electrodes surface, without the need to remove any edge shielding. A basic clean is required if arcs are being generated/observed on the electrode in racetrack, or the area inside the racetrack.

If the contamination is found to be loose process particles deposited during a venting cycle or film pieces deposited during a web break, an IPA/acetone clean with a lint free cloth is sufficient.

Figure 8.1-23 Example of an operator carrying out initial cleaning with isopropanol alcohol and a lint-free cloth.

Procedure:

1. Open the copper front covers and seal the outer edges of the plates as shown below.
2. Use a form of masking tape to seal around the electrode plates to prevent debris and dust particles falling down the edges of the plasma treater.

3. Use a fine grade abrasive paper to remove any deposits or surface imperfections by hand.
4. Gradually move to finer grades of abrasive material to remove any scratch marks.
5. Following the removal of the contamination, thoroughly vacuum all surfaces of the electrode, followed by a final full clean with alcohol or IPA solution.

Deep Clean

Definition: A “deep clean” refers to using a machine such as a disk or orbital sander fitted with a fine grit to remove surface imperfections or deposition build-up.

If a hard, abrasive deposit is found adhered to the plates from a deposit from the film, or AlOx formed at the edge of the racetrack during prolonged process running of the plasma treater, a through clean is required.

Procedure:

1. Open the copper front covers and seal the outer edges of the plates as shown below.
2. Use a form of masking tape to seal around the electrode plates to prevent debris and dust particles falling down the edges of the plasma treater.
3. Use a sanding machine to remove any deposits or surface imperfections.
4. Gradually move to finer grades of abrasive material to remove any scratch marks evident on the plates.
5. Following the removal of the contamination, thoroughly vacuum all surfaces of the electrode, followed by a final full clean with alcohol or IPA solution.

Figure 8.1-24 Electrode plates with masking around the edges.

Figure 8.1-25 A machine such as an orbital sander can be used so long as it is fitted with fine grit abrasive material.

	Note !
	It is suggested to use a minimum of 250-grade abrasive material, ideally 1000 grade.

8.2 Cleaning Schedule

The following is intended as a guide only and generally users will find their own sequence of cleaning based upon their particular operating methods and time periods.

	Caution !
	**Aluminium oxide debris/waste materials after coating forms fine powders and flakes, Personal Protective Equipment (PPE) must be worn when cleaning machine (Rated goggles & face mask, etc).** **With the increased risk of fire being present, suitable clothing must be used while operating this equipment. Fireproof overalls and thermal gloves, etc** **In regards to disposal of waste, non-flammable storage measures must be used, as described within this manual**

Any solid remaining aluminium debris must be collected and stored using suitable non-flammable fire-proof containers, as described in this document. These containers MUST be clearly identified for the purpose, either by graphics/wording and or colour coded.

	Warning !
	**No other types waste materials should be placed in these designated waste storage drums, as *potentially large amounts of waste aluminium oxide present a potential fire hazard* if it comes into contact with water.**

Cleaning Between Runs

This procedure assumes that all other processes have been competed, and debris has been removed from the machine.

1. Visually check the evaporation source area and remove any loose debris from coating shield.
2. Visually check the aluminium wire feed area for any potential wire feed “snags/crossovers”.
3. Check that there is sufficient aluminium wire on the spools for the next process cycle.
4. Use a used evaporator or a scraper to clean oxide debris off the evaporators.
5. Check and clean the aluminium wire feed spouts (tubes) as required.
6. Check the chamber joint band and 'O' ring for obstructing debris.
7. Dry the cryogenic coils and walls with a limit free cloth.
8. Remove any loose substrate debris

Shift Clean Up (To be performed at the end of each shift)

1. All the above plus the following.
2. Inspect the coating shields, remove any build up and apply release solution.
3. Check the rubber covered rollers for aluminium build up and surface slickness.
4. Remove any loose substrate debris in the machine.

Weekly Clean Up

The following clean up tasks should be performed on a weekly basis.

Evaporation Source Area

1. Clean evaporation source box “back” to bare metal.
2. Clean copper evaporator posts and blocks to bare copper.
3. Remove wall shields (if fitted) and clean.
4. Clean top and bottom of movable shutter.
5. Wipe all supports down.
6. Aluminium wire feed tubes to be cleaned or replaced.

Coating Area

1. Clean all shielding down to original surface.
2. Wipe all surface areas.
3. Wipe zone sealing strip with clean cloth.
4. Wipe main chamber 'O' ring and lubricate it
5. Check zone separation gap around the process drum is clear of debris.
6. Remove all aluminium debris from above the zone separation seal.

Winding Mechanism Area

1. Clean all rollers as described in the previous section of this manual.
2. Remove any debris build up.
3. Clean any oil spillage etc.
4. Check for substrate trapped within the inline detection system, and remove.

Viewing Windows

1. Clean all viewing windows inside using dry cloth.

Plasma Units

1. Clean electrode plates and remove any debris

General Cleaning

1. Clean all external machine surfaces.
2. Clean loading area and metallizing room.
3. Empty all waste and scrap bins.
4. Empty vacuum cleaner.

The above schedule is provided to be used as a guide.

Individual customers may find that they will have to modify this schedule depending upon their operational experiences. However, all necessary measures should not be compromised.

8.3 Disposal of Waste Materials

This chapter describes the waste disposal methods for the consumables used by the machine.

General Notes

After dismantling any sections of the machine, for disposal, the components and process media must be separated in the following categories:

- Process media.
- Components contaminated with process media or which had contact with process media.
- Other machine components.

	**Note !** **Components and process media must be disposed of in accordance with local site regulations where the machine is operated.**

Process Media

	**Warning !** **Pump oil and oil mists contaminated with hazardous substances can be hazardous to your health.** **Skin or eye contact with pump oil or inhaling oil mists can be hazardous to your health.** **Wear protective clothing, oil-resistant gloves, goggles and a P3 respiratory protection mask when opening vacuum lines, vacuum valves or vacuum pumps containing harmful gases or other harmful substances, and when disposing of exhaust condensation or filters.**

	**Caution !** **The process media of the vacuum pumps are harmful to the environment if not disposed of correctly.** **Never allow pump oil or parts dipped in pump oil to penetrate the supply of drinking water. Pump oil must be disposed of as hazardous waste.**

Process media for the machine should be returned to the supplier in unopened original containers.

Process media in opened original containers and used must be disposed of as hazardous waste in accordance with local regulations.

Consumables items containing process media (filter cartridges, cleaning rags, etc.) must also be disposed of as hazardous waste.

Components

All machine components must be separated in the following classes, and recycled or disposed of, depending on the material:

- Aluminium
- Steel
- Stainless steel
- Precious metal
- Copper
- Glass

- Plastic
- Electronic or electrical parts
- Batteries

Components contaminated with operating media must be thoroughly cleaned before further processing.

Plasma Treater Target Materials

Precious metal target materials can be recycled.

Other target materials must be disposed of in accordance with the safety data sheets and local regulations.

Aluminium and Aluminium Oxide Waste Materials

Solid remaining aluminium or aluminium oxide debris must be collected and stored using suitable non-flammable fire-proof containers that can be sealed against water ingress. These containers MUST be clearly identified for the purpose, either by graphics/wording and or colour coded.

Warning !

No other types waste materials should be placed in these designated waste storage drums, as *potentially large amounts of waste aluminium oxide present a potential fire hazard* if it comes into contact with water.

Disposal of these materials should be made through a local approved and registered waste disposal company.

8.4 Process Drum Cleaning

In the situation where the process drum becomes coated with evaporated aluminium, for example when a web break occurs, it has to be cleaned of any aluminium that builds up on it.

The drum can be pushed around manually, or can be driven around using the drive system.

Rotation of the Process Drum

The drive system has been designed to allow rotation of the process drum in a controlled manner by the machine operator. To facilitate this, two devices can be supplied by BOBST

- A foot switch is supplied, with a cable that is plugged into a socket on the winding cart panel (E0_6). This will allow the drum to rotate at a speed of 20 Metres/min
- An optional cleaning device that plugs into the same socket, but allows the drum to operate at a high speed of 280 m/min. This device incorporates a cleaning system, attachments to a vacuum system and a higher level of safety, due to the high speed of rotation.

Warning !

The footswitch switch should never be overridden by a mechanical obstruction.

Figure 8.4-1 Example of an operator using the foot switch

Direction of Drum Movement

The drum will operate at different speeds and directions dependent upon device used.

- Footswitch: Drum rotates in normal machine direction at a speed of 20 metres/min

Note !

There are various machine interlocks to stop the machine closing if the activation switch is not removed from the socket.

SECTION 9.0 AUXILIARY EQUIPMENT

The following section contains information about other machine features and functions that are supplied within the scope of the machine delivered.

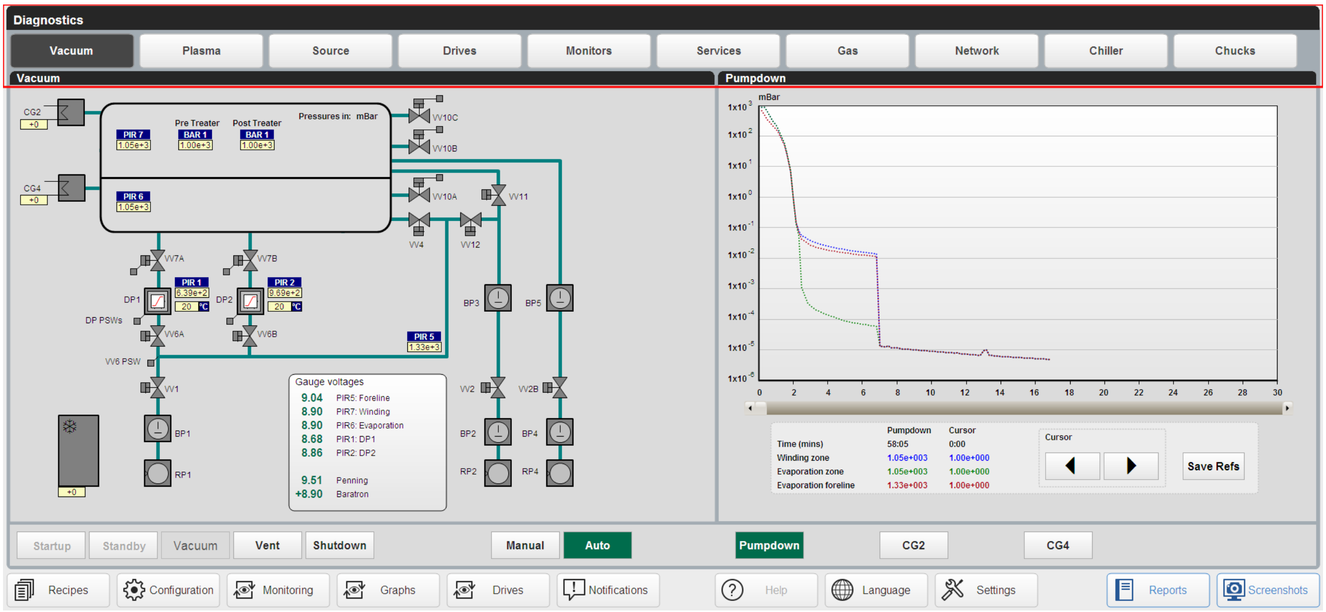
9.1 HMI Diagnostic Interface

The user of the machine can check on the status of a number of machine sub sections through the diagnostics HMI screen.

Press the settings button to open the settings menu and press diagnostics to view the diagnostics page.

Figure 9.1-1 HMI DIAGNOSTIC screens access method

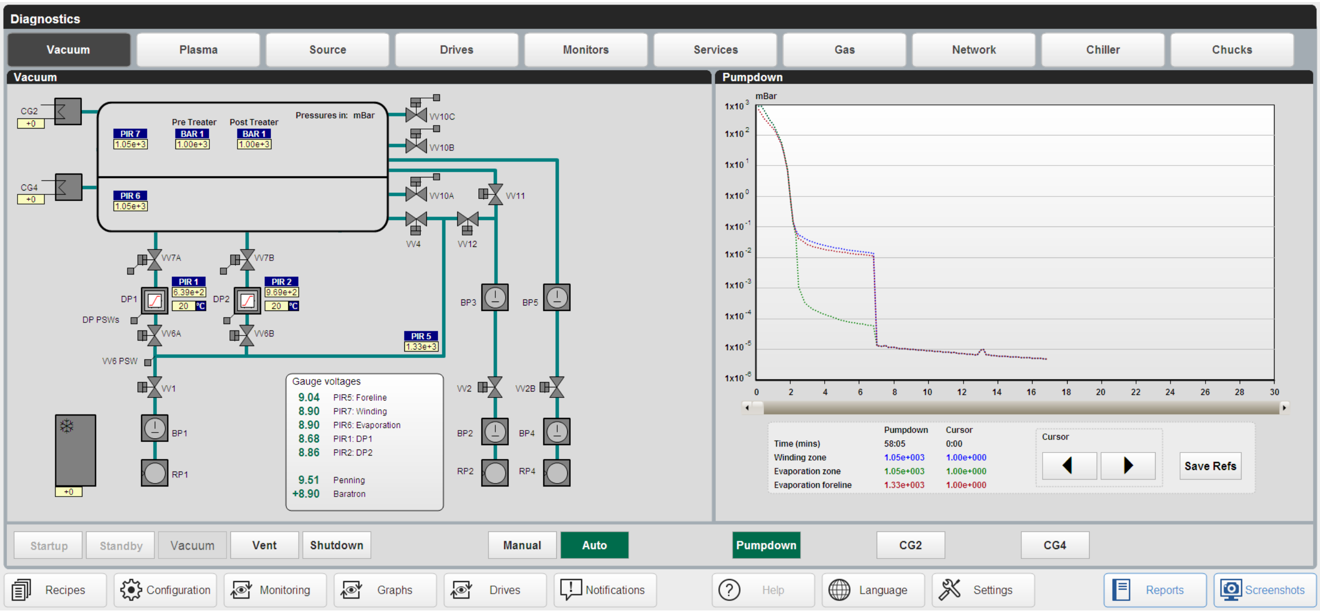
The diagnostics page is split into multiple sections that represent different features on the machine:



Tap a section to view the diagnostics for each feature.

Vacuum Diagnostic HMI Screen

The vacuum screen has two sections, vacuum and pumpdown.



The vacuum mimic section is displayed on the left side:

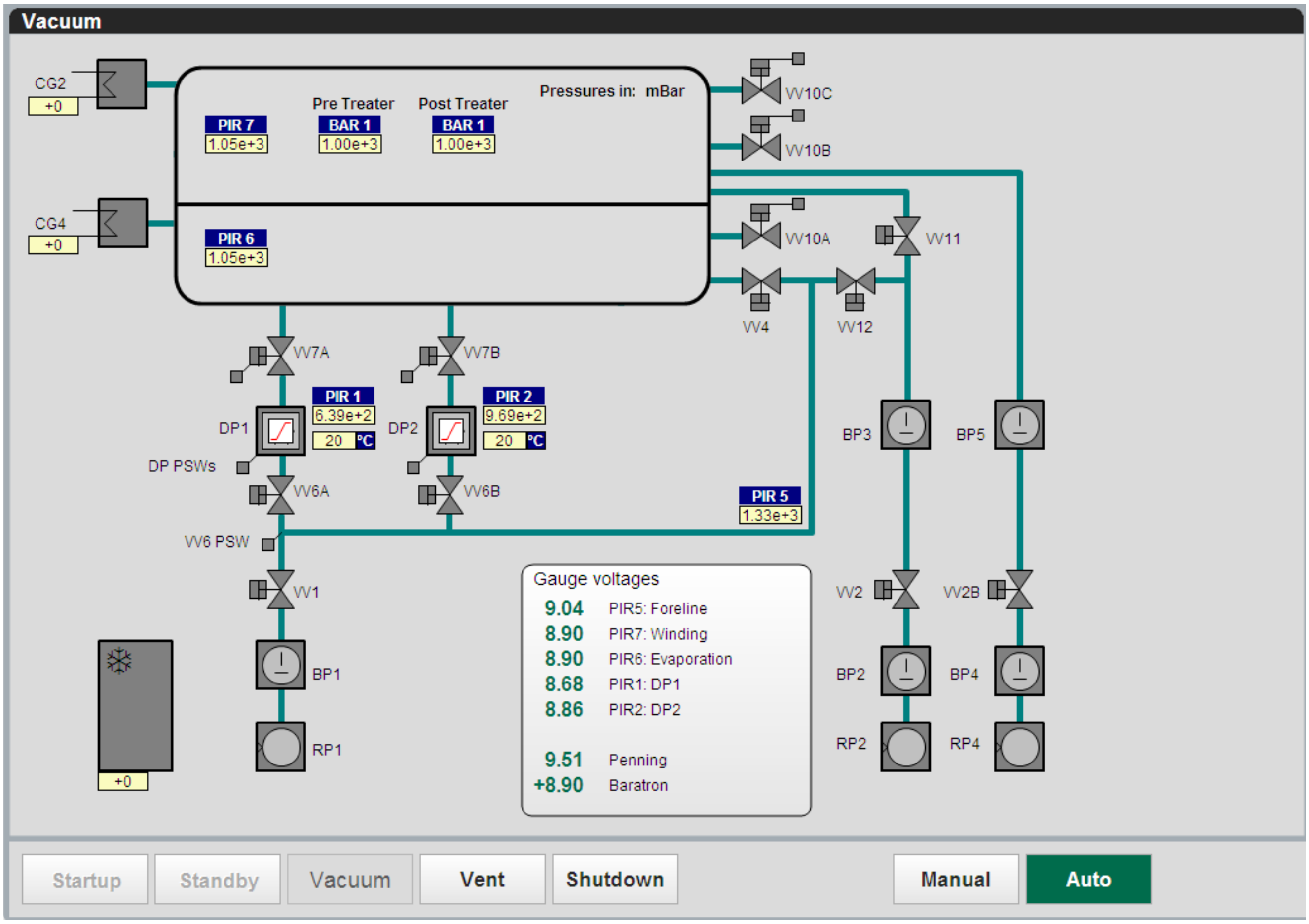


Figure 9.1-2 VACUUM diagnostic HMI screen – operation mimic

The pumpdown and polycold sections are displayed on the right:

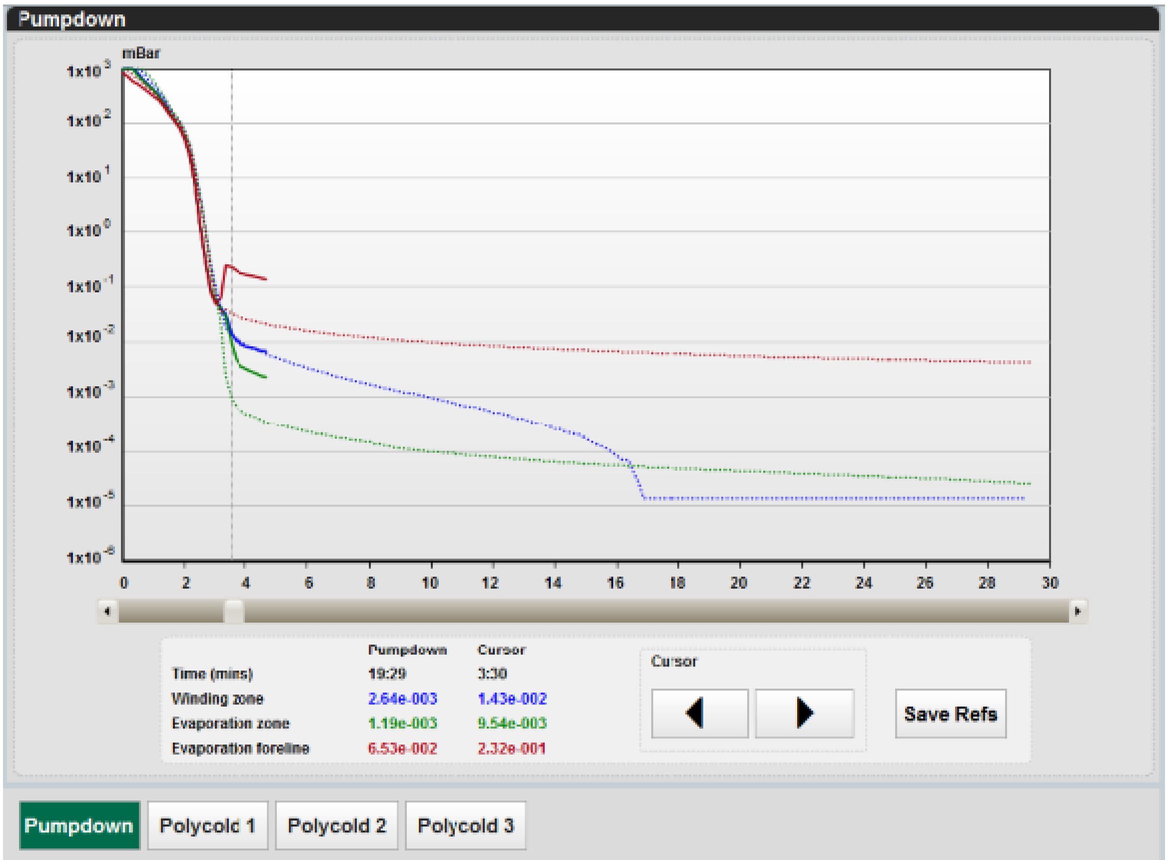


Figure 9.1-3 VACUUM diagnostic HMI screen - pumpdown

The pumpdown section allows the user to monitor and compare the evacuation of the chamber every process cycle.

Tap a cryogerenerator or polycold option to view its status:

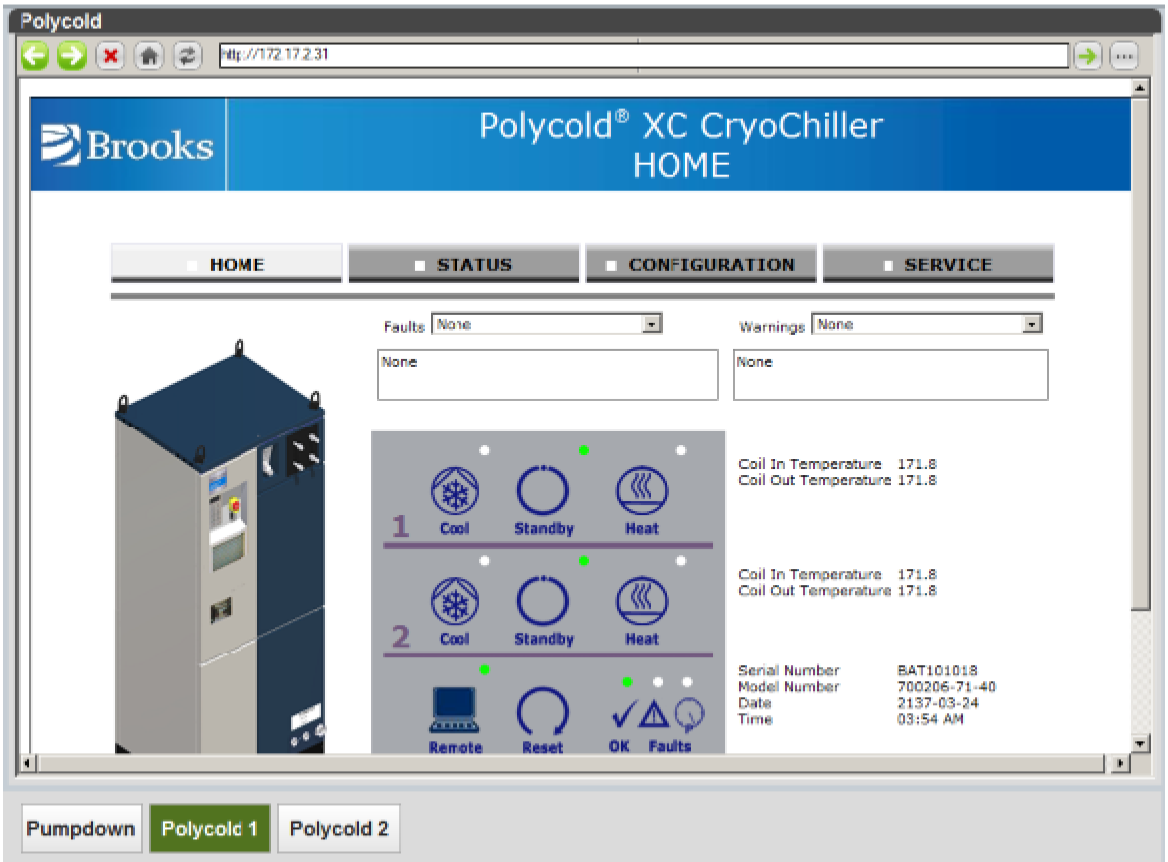


Figure 9.1-4 VACUUM diagnostic HMI screen – Polycold cryogenerator graphical user interface (GUI)

Plasma Diagnostic HMI Screen

The plasma screen shows pre-treater mimic screen.

Figure 9.1-5 PLASMA diagnostic HMI screen

The pre-treater mimic screen is displayed on the left.

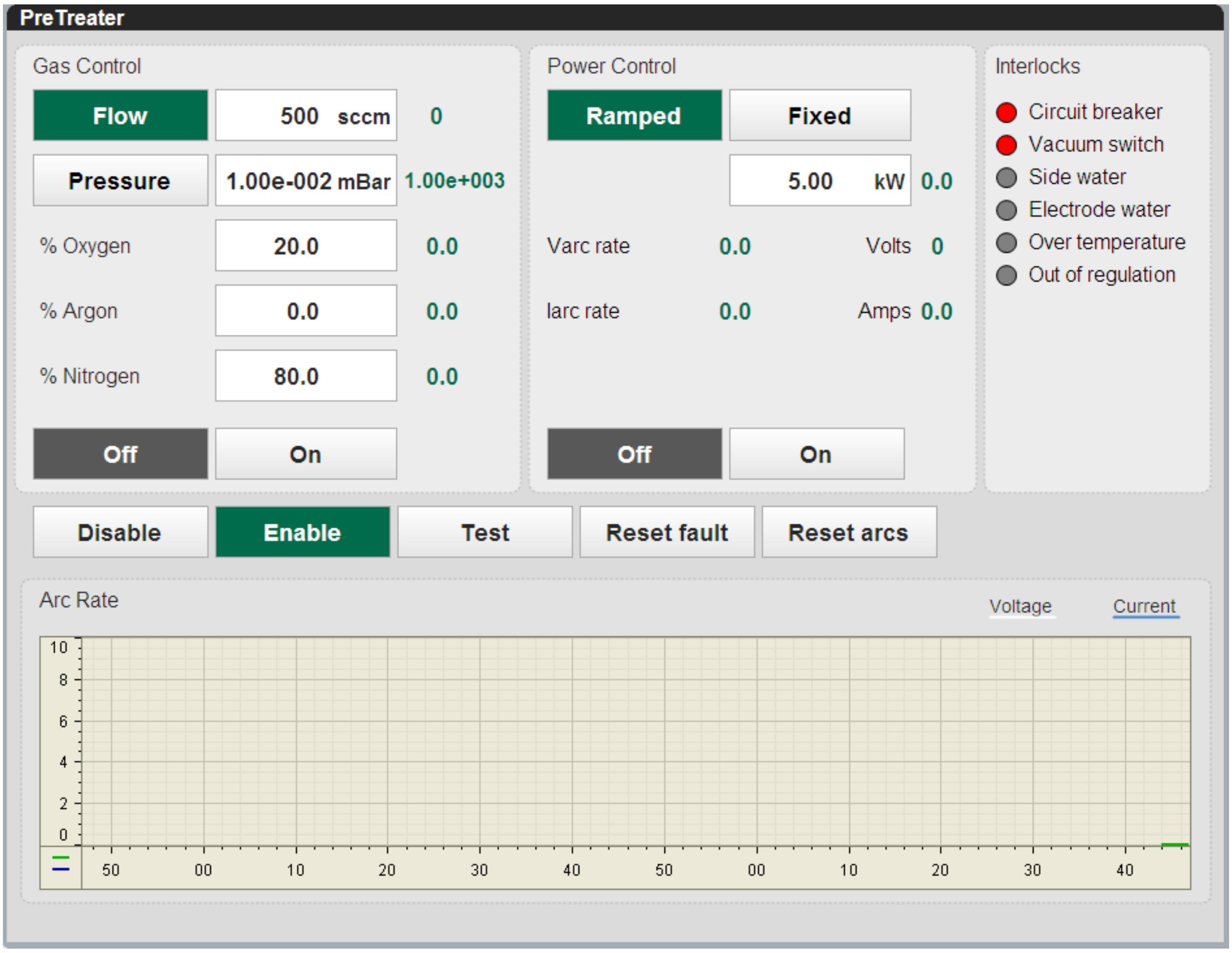


Figure 9.1-6 PLASMA diagnostic HMI screen, pre-treater

	**Note !**
	Explanations of the information displayed on these screens will be given in section 8 of this manual

Source Diagnostic HMI Screen

The source diagnostics page shows information about the thyristors, wire feeders and source thermocouples.

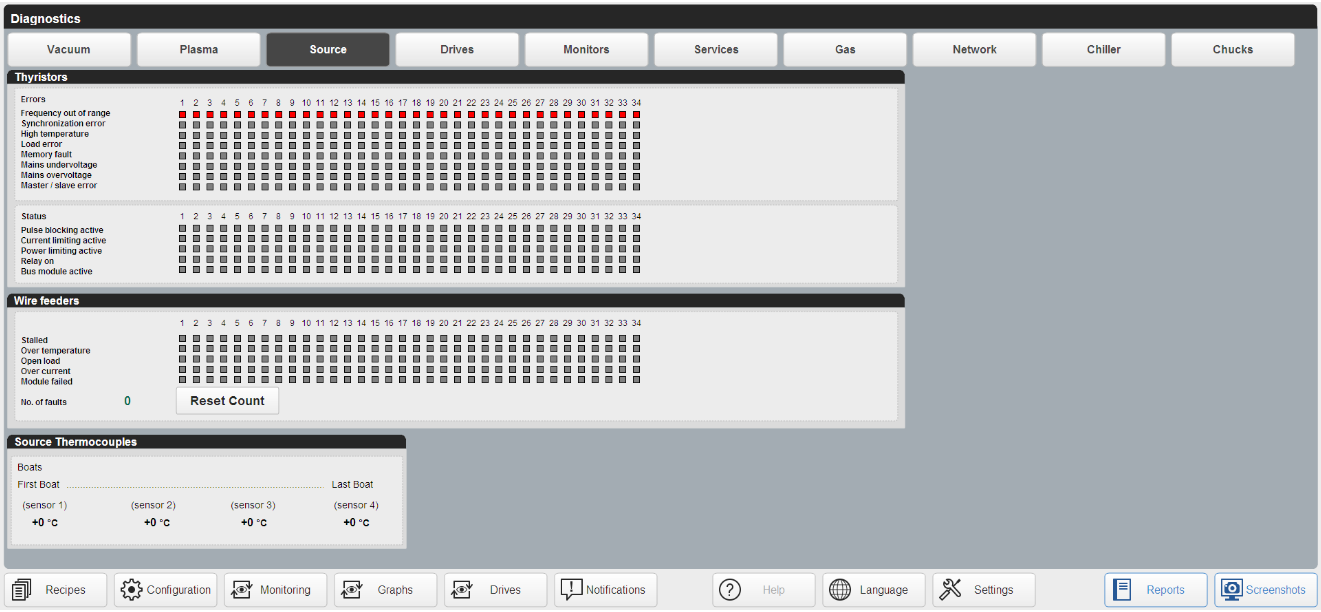


Figure 9.1-7 SOURCE HMI diagnostic screen

This thyristors section allows you to monitor the power supply thyristor units of each evaporator.

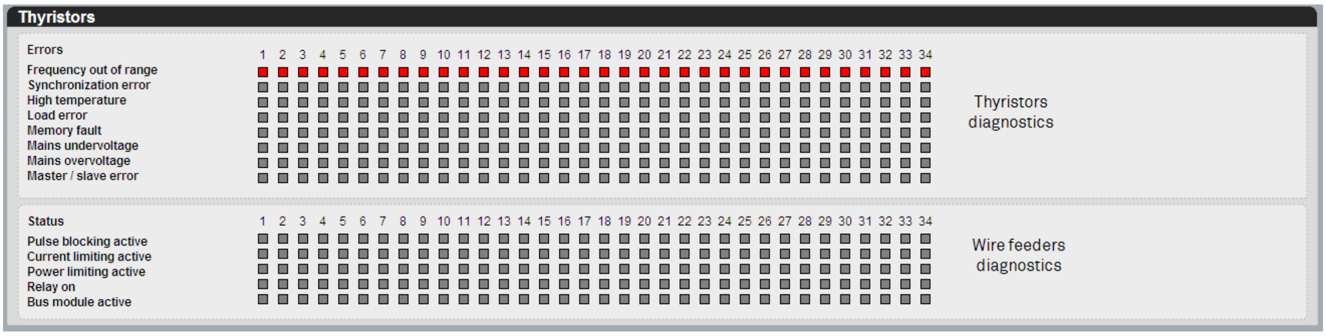


Figure 9.1-8 SOURCE HMI diagnostic screen - thyristors

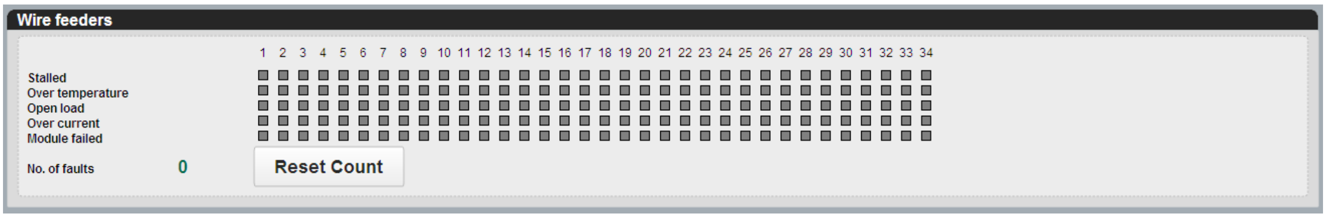
A line under voltage message is displayed when the boats off condition is monitored.

Error Indication	Description
Frequency out of range	Outside pre-defined limits of 47-63 Hz
Synchronization Error	Device recognises that there is no-Zero crossing within the gate
High Temperature	Internal temperature too high
Load Error	Open circuit identified.
Memory Fault	N/A
Mains Undervoltage	Mains supply outside preconfigured limits of mains supply -10%
Mains Overvoltage	Mains supply outside preconfigured limits of mains supply +10%
Master/Slave Error	Issue with control cabling on dual systems. Not applicable to all machines

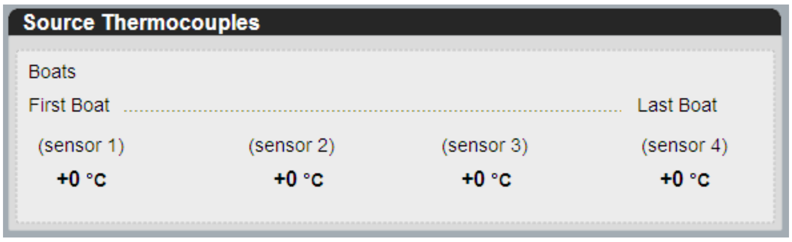
Table 9.1-1 Thyristor stack diagnostic error indicators

Status Indication	Description
Pulse Blocking Active	Indicates that the unit is not being enabled and the firing pulses have been blocked
Current Limiting Active	Indicates that the unit is limited by maximum current being drawn from the unit.
Power Limiting Active	Indicates that the unit is limited by maximum power being drawn from the unit.
Relay On	Indicates that internal relay is active to close inputs X1.5 & X8.5. (Not currently used by BML)
Bus Module Active	Indicates that the bus module is operational

Table 9.1-2 Thyristor stack diagnostic status indicators



The wire feeder status messages are described by the wording.



Drives System HMI Diagnostic Screen

This drives screen gives the status of the winding system drive components, and the individual aluminium wire feeder controls on the machine. Please refer to the vendor manuals for issues with these components.

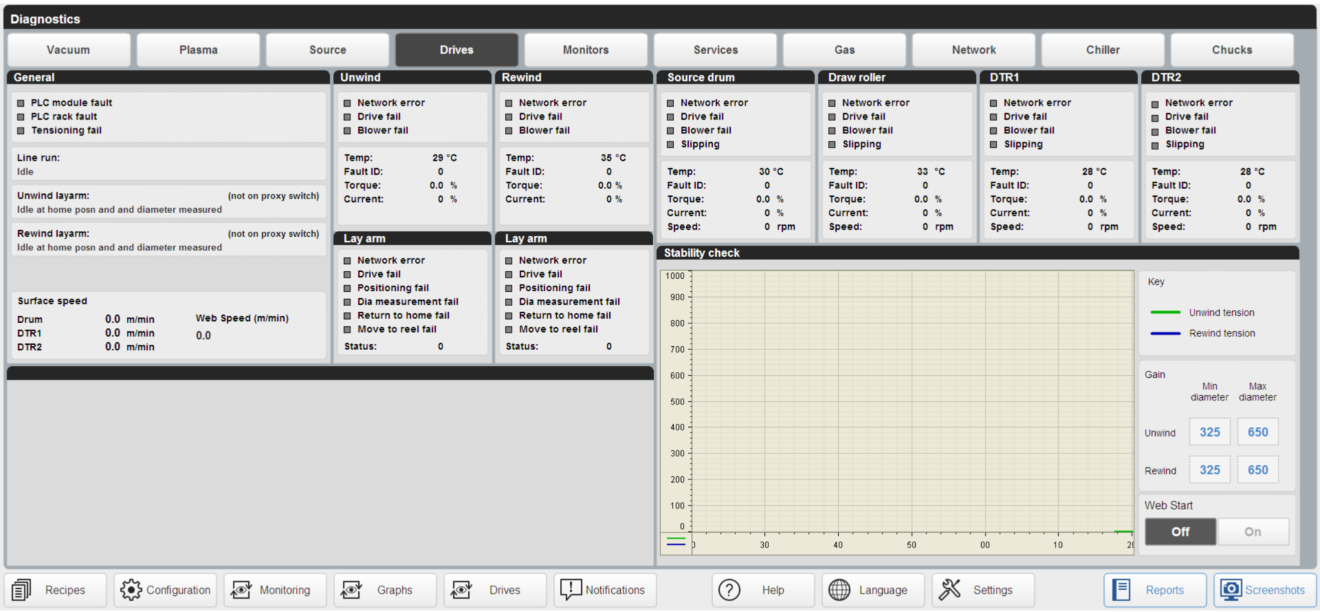


Figure 9.1-9 Drives HMI diagnostic screen

The general section will show a general information about the current state of the drives on the machine:

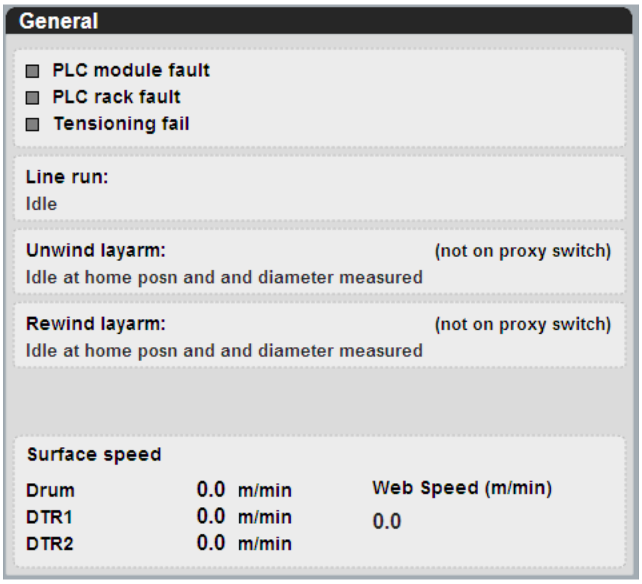


Figure 9.1-10 Example of the general information section on the drives diagnostics HMI screen

Each other section shows information that is specific to that drive system:

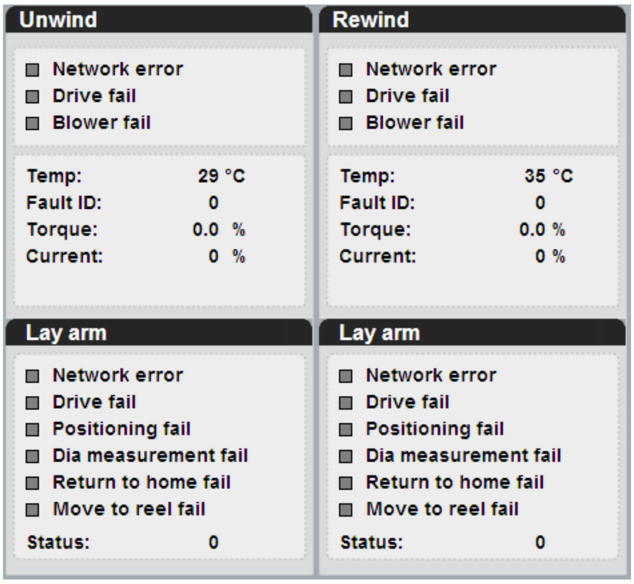


Figure 9.1-11 Example of the unwind and rewind section on the drives diagnostics HMI screen

An error box will be highlighted when the machine registers that error on the drives system.

The fault ID and layarm status count default is zero. A code is displayed when the drive system has an alarm condition. Use this code to refer to the vendor documentation for the reason for the alarm.

Note !

In the case of a drive system fault, the service technician will ask for the code displayed.

See chapter 9.3 HMI Status Messages on page 415 for details. of the messages displayed.

Monitor HMI Diagnostic Screen

This screen is used to troubleshot/set up the deposition monitor and closed loop control system.

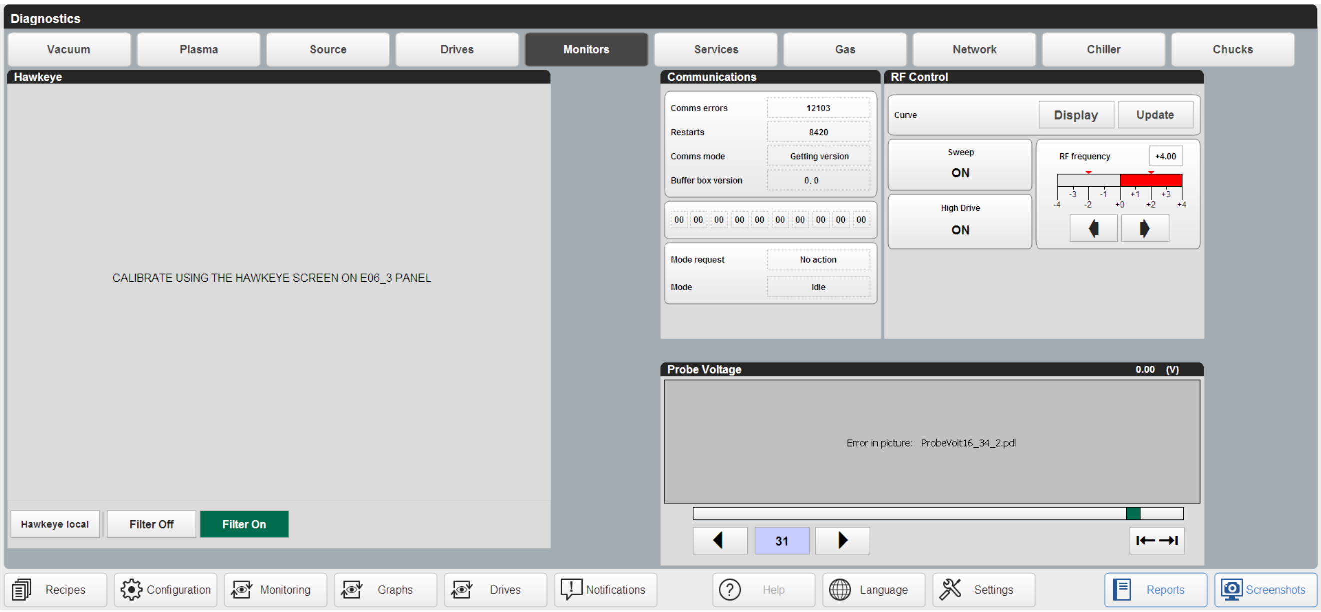


Figure 9.1-12 Deposition monitor HMI diagnostic screen

	Note !
	Do not adjust the values in the monitor tuning section unless given direct instructions from Bobst Manchester Ltd, as any made may affect the closed loop operation of the machine.

Bobst Manchester Ltd technical support may ask for information in this screen if there are issues with the deposition system control.

Services HMI Diagnostic Screen

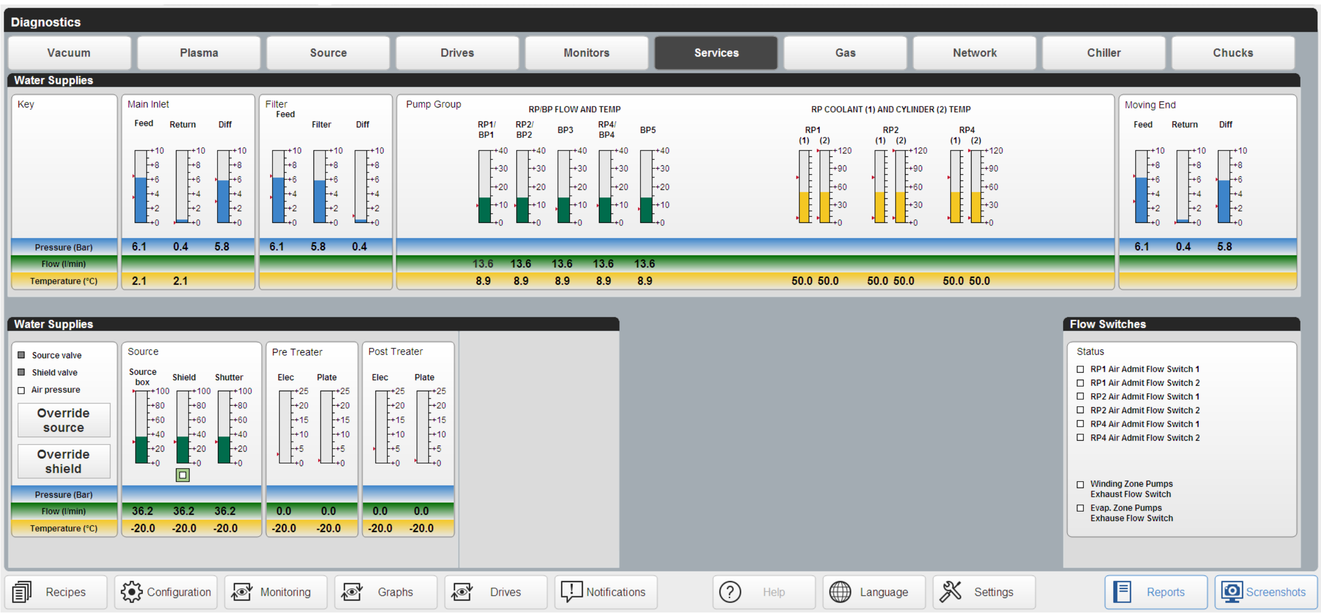


Figure 9.1-13 Services HMI diagnostic screen

This screen gives the status of various electrical panel components and general systems on the machine.

Pressure monitoring

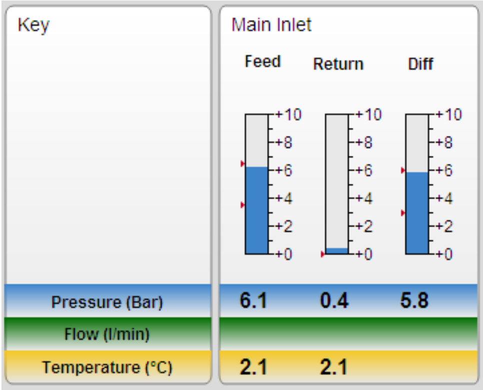


Figure 9.1-14 Water pressure diagnostic section of the screen

The main inlet section has three indicators.

- Inlet (Feed)
- Outlet (Return)
- Differential.(Diff)

The red arrowheads on the outside of the columns indicates the pre-set operating levels that the system is to work within.

The blue line below the chart shows the pressure.

Flow Monitoring

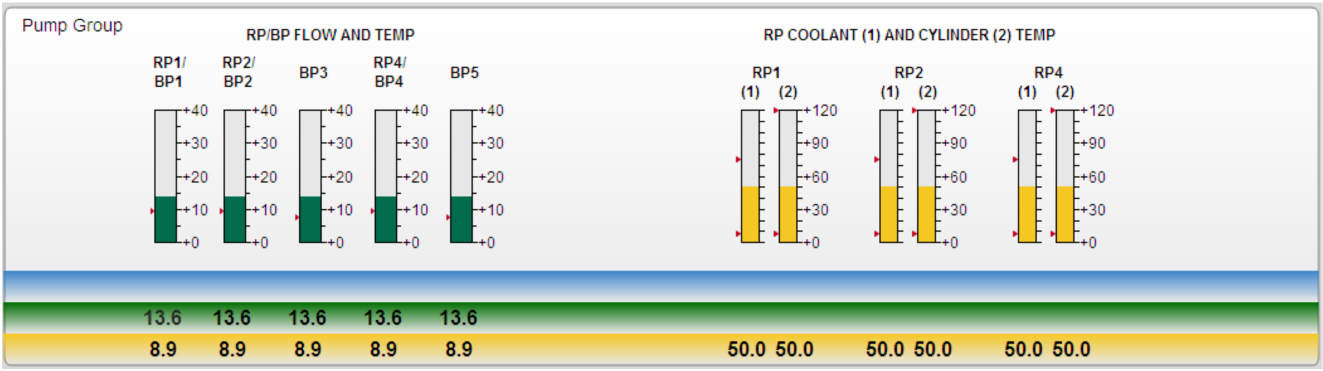


Figure 9.1-15 Water flow diagnostic section of the screen

There are flow monitors on critical circuits. The outputs are displayed in the pump group section. For example RP/BP (Vacuum booster pump).

The yellow line below the chart shows the temperature. This is the temperature of the water passing through the measuring device.

If the water supply to the machine reaches the limits of +28/+22 degrees, a warning will be given. If the temperature continues to increase to +30 centigrade, an out of limits alarm will be given and most of the machine systems will close down.

Temperature Monitoring – Incoming supply

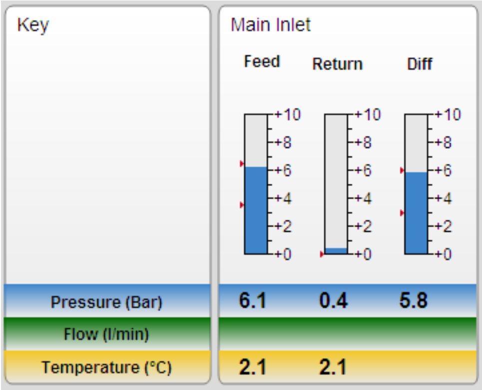


Figure 9.1-16 Incoming water diagnostic section of the screen

The incoming water supply temperature to the machine is monitored to ensure that equipment is not operated outside the OEM recommended requirements.

The red arrows on the left-hand side of the display indicate these limits. The display will change to the colour red outside these limits and an alarm condition will be notified.

The yellow line below the chart shows the temperature.

Temperature Monitoring – Rotary Pump Coolant

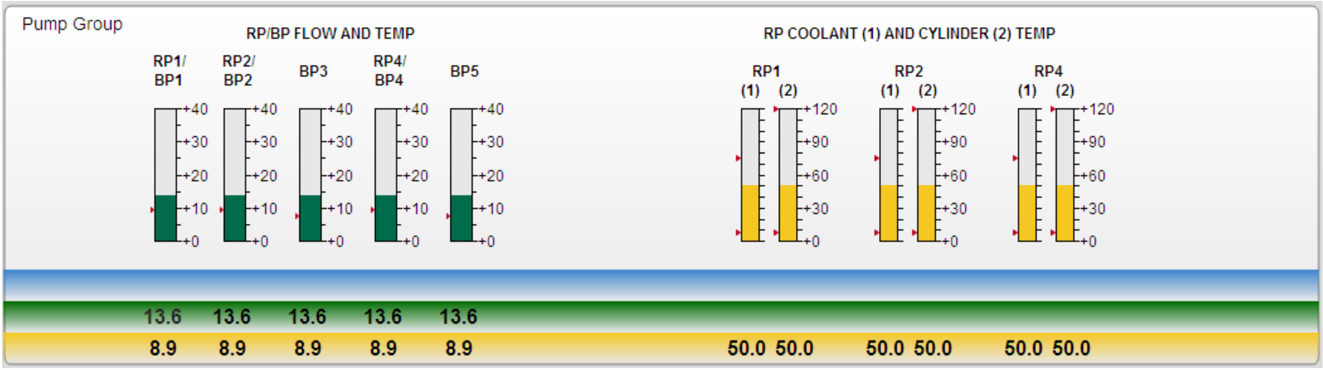


Figure 9.1-17 Rotary pump temperature monitoring

The temperature of the coolant contained within the pump body is indicated in this part of the screen.

The red arrows on the left-hand side of the display indicate these limits. The display will change to the colour red outside these limits and an alarm condition will be notified.

The yellow line below the chart shows the temperature.

Isolation Valve Operation

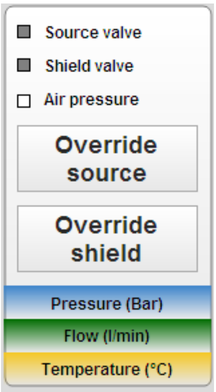


Figure 9.1-18 SERVICES diagnostic screen – Water Isolation

To prevent condensation forming on water cooled components in the vacuum environment, the circuits are isolated automatically.

This section of the screen allows the valves to be opened, by overriding them, using the buttons provided.

This allows circuits to be checked for leaks, etc.

Gas HMI Diagnostics Screen

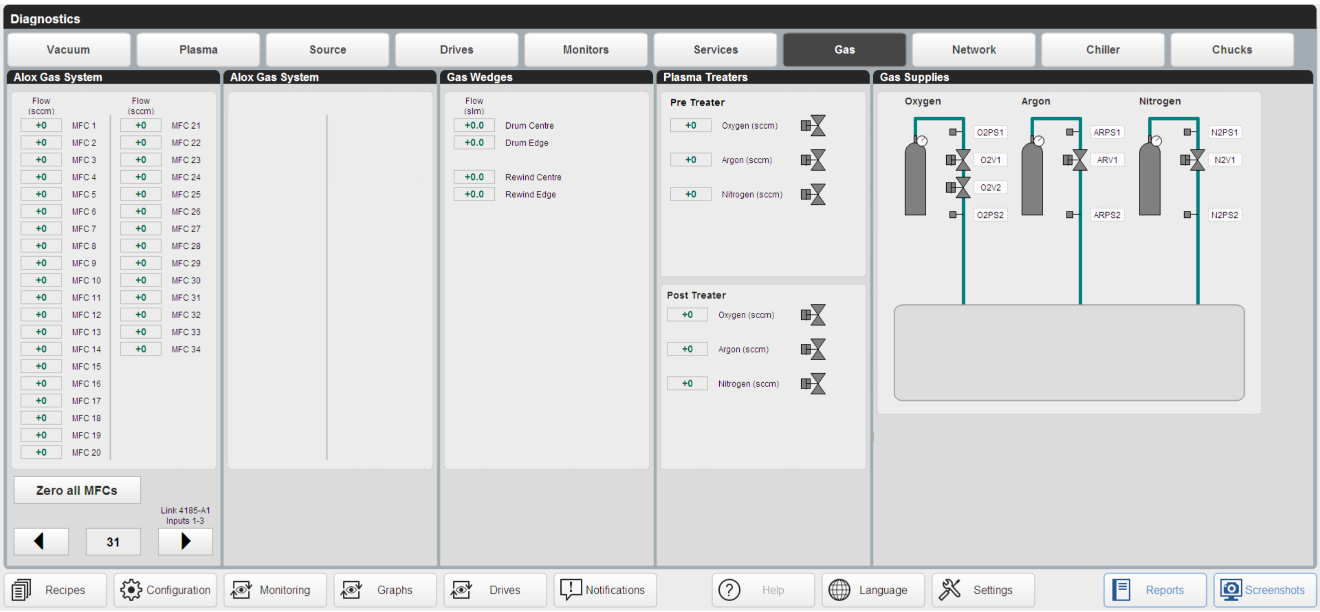


Figure 9.1-19 Gas HMI diagnostic screen

Network HMI Diagnostic Screen

This screen shows the status of the control items located around the machine that are linked on Asi and profibus networks.

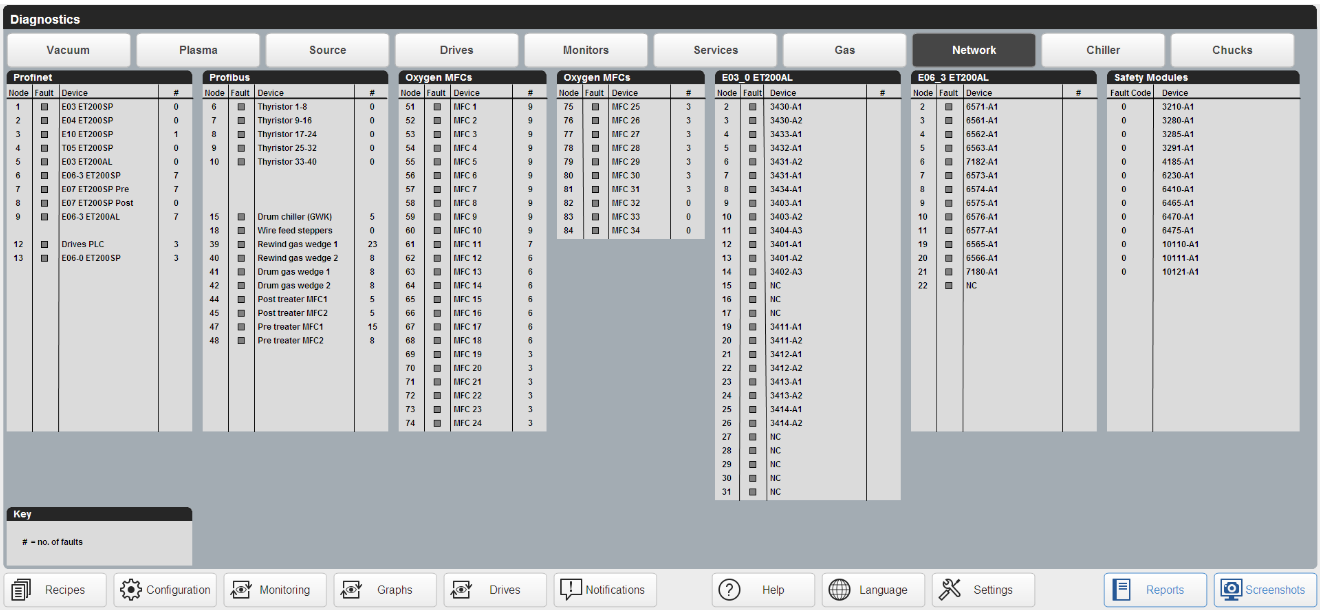


Figure 9.1-20 NETWORKS HMI diagnostic screen

The status of the individual nodes are colour coded;-

- Grey: - Network element/node is monitored and is operational mode.
- Red: - Fault or not connected, or powered up.

In the situation where some units are powered down in the case of a E-stop being activated, once power has been reapplied, the 'RESET I/O' button has to be touched

Chiller HMI Diagnostic Screen

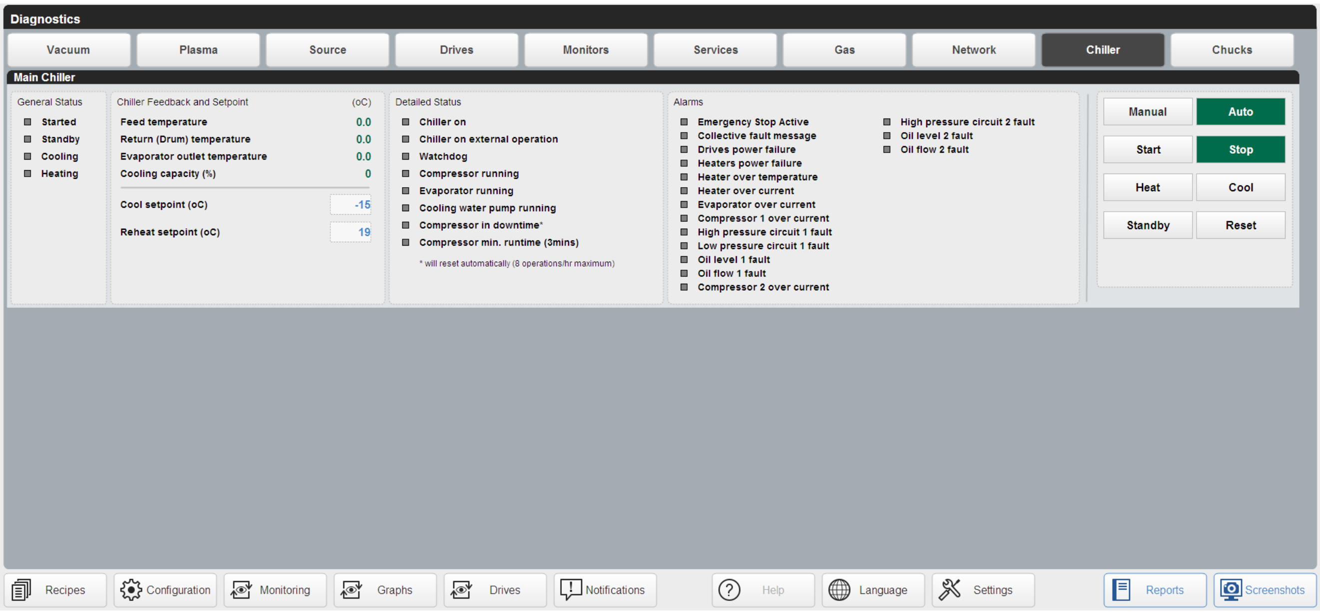


Figure 9.1-21 Chiller HMI diagnostic screen

Button	Description
Manual	Selects manual chiller operation.
Auto	Automatic chiller operation. The chiller will change from cooling to heating at the end of the coating cycle. The cooling mode is selected when the vacuum pump-down sequence is started.
Start	Switch the chiller on. Can only be used when the chiller is in manual mode.
Stop	Switch the chiller off. Can only be used when the chiller is in manual mode.
Heat	Select heating mode. Can only be used when the chiller is in manual mode.
Cool	Select cooling mode. Can only be used when the chiller is in manual mode.
Standby	Select standby mode. Can only be used when the chiller is in manual mode.
Reset	Reset any faults to allow the chiller to restart.
Prime pumps	Causes coolant circulation within the chiller. The process feed and return valves are closed. Only available at Maintenance Level.
Maintenance pumpdown	Allows the chiller to be operated independently of the rest of the machine. Only available at Maintenance Level.

Table 9.1-3 Chiller diagnostic buttons

Chucks HMI Diagnostic Screen

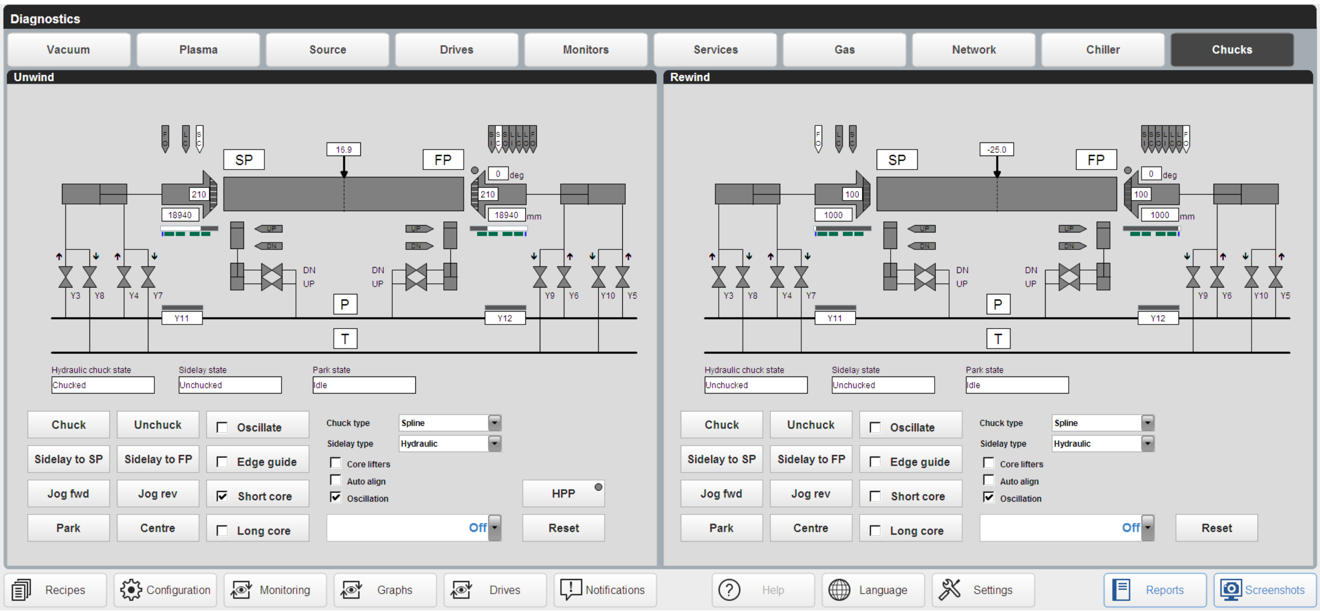


Figure 9.1-22 Chucks HMI diagnostic screen

9.2 HMI Generated Alarms and Warnings

The machine will report notifications that occur with the machine hardware and services supplied to the machine.

Notifications are displayed in a box at the top of the screen.

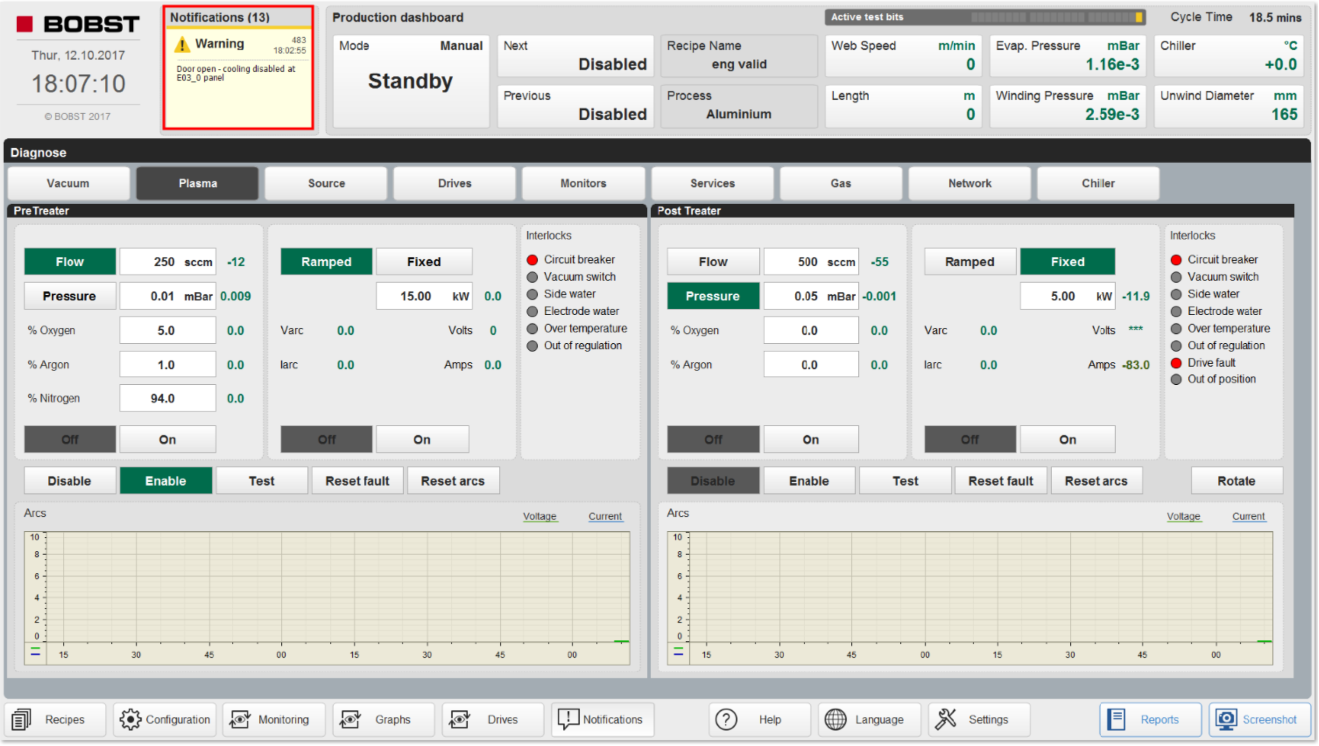


Figure 9.2-1 Screen capture of a screen, indicating NOTIFICATION on the screen

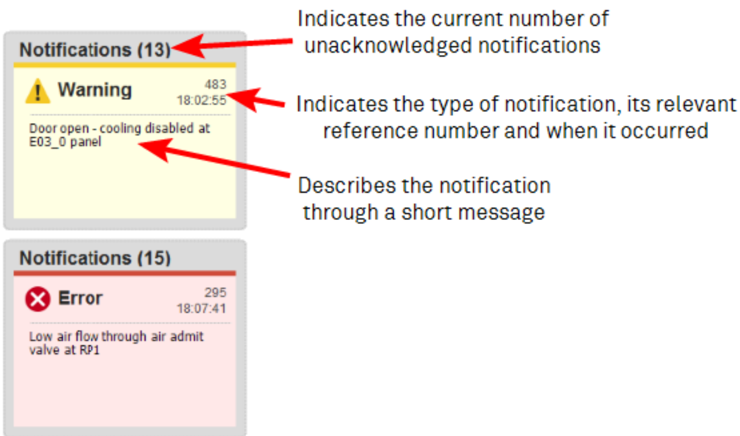
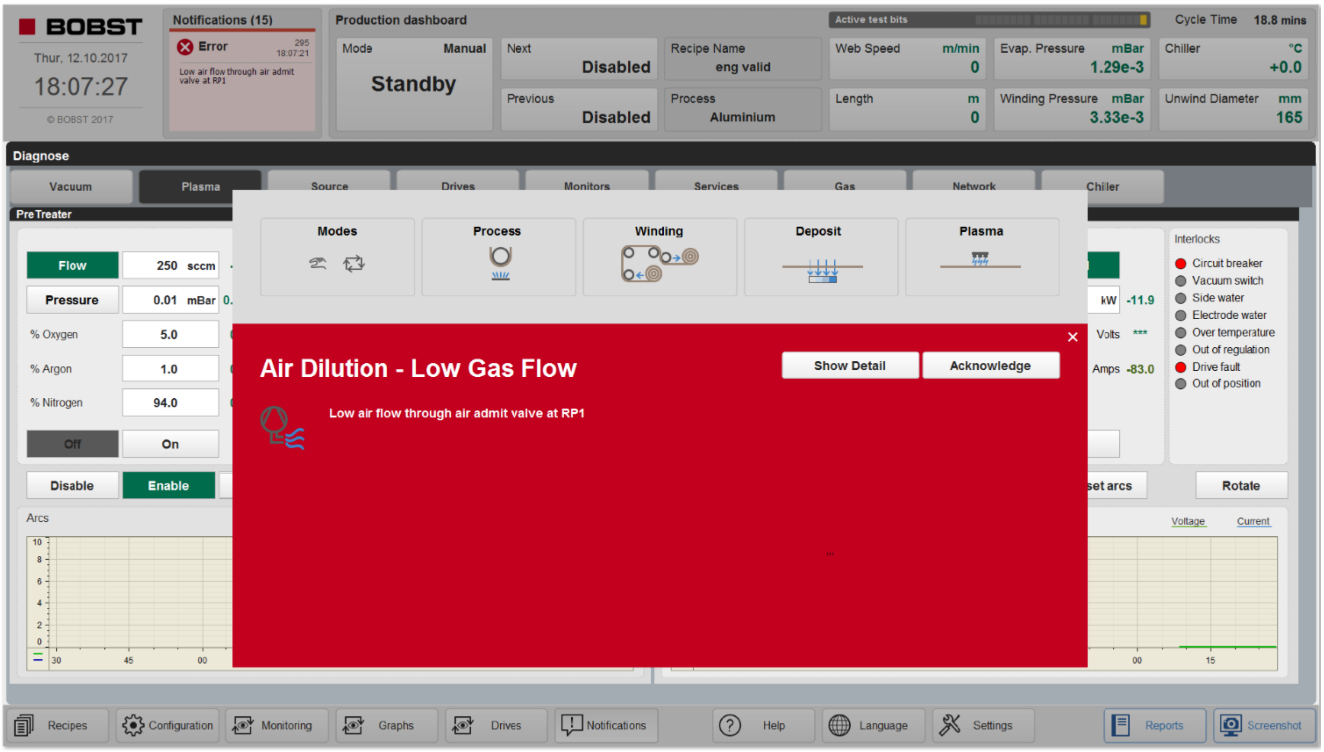


Figure 9.2-2 Screen capture of typical notification messages

- Touch the notifications section to open the notifications screen.

Errors are also displayed in a popup:

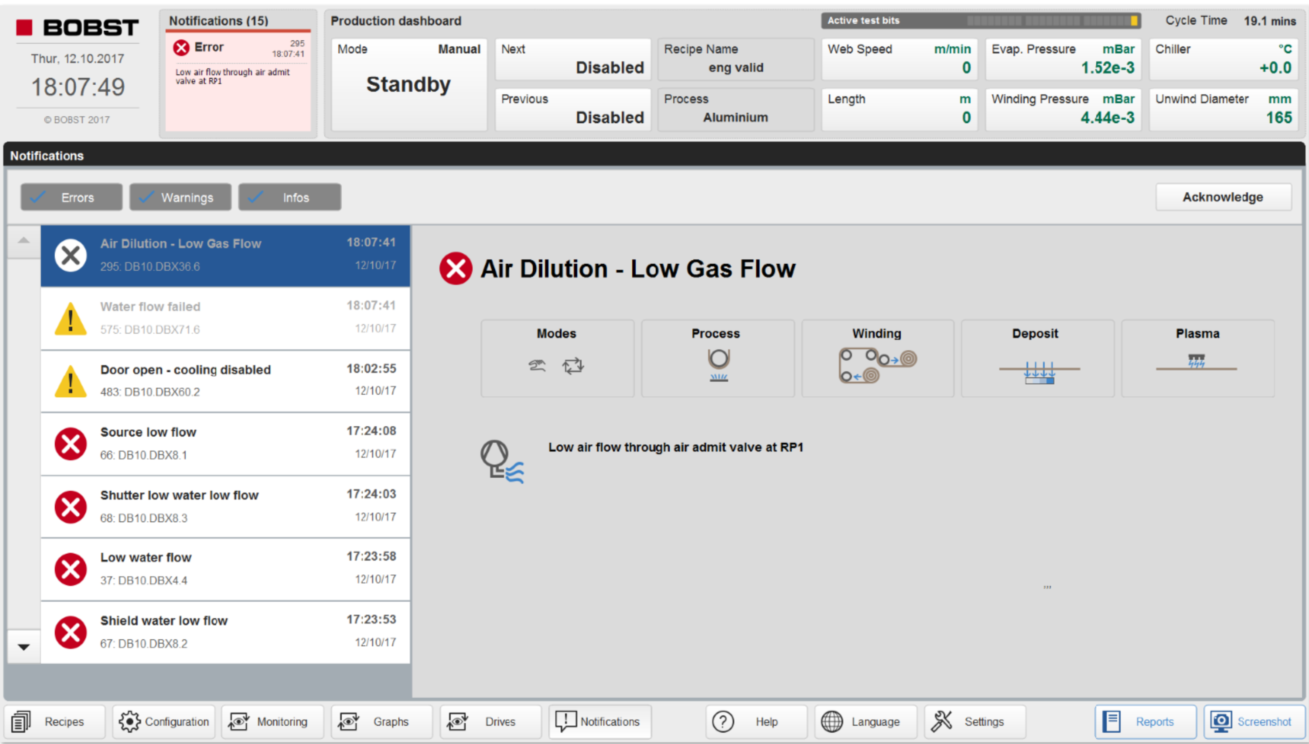


The icons at the top of the popup will highlight where the error has occurred. Errors require action from a user to resolve an urgent issue.

- Touch Acknowledge to dismiss the notification
- Touch Show Detail to view the notification in the notifications page

Figure 9.2-3 Screen Capture of HMI NOTIFICATION screen

Notifications Page



Notifications are categorised into three types:

- Errors notify the user of an urgent issue that needs to be resolved. Touch the acknowledge button to dismiss the notification when the issue is resolved.
- Warnings notify the user that an action cannot be performed due and why. Touch the acknowledge button to dismiss the notification.
- Infos notify the user of any other information. Touch the acknowledge button to dismiss the notification.

Alarm Reference

Note !

Not all these messages apply to all variants of this type of machine. It will depend upon the optional equipment that is installed.

Warning or Error Number	Short Description	Long Description
1	Valve fault - VV1	Valve fault - Vacuum Valve VV1:Please seek technical assistance
2	Safety system fault	Check Diagnostics | Network screen for fault code
3	Valve fault - VV2	Valve fault - Vacuum Valve VV2:Please seek technical assistance
4	Valve fault - VV2B	Valve fault - Vacuum Valve VV2B:Please seek technical assistance
6	Valve fault - VV4	Valve fault - Vacuum Valve VV4:Please seek technical assistance
8	Valve fault - VV6A	Valve fault - Vacuum Valve VV6A:Please seek technical assistance
9	Valve fault - VV6B	Valve fault - Vacuum Valve VV6B:Please seek technical assistance
10	Valve fault - VV7A	Valve fault - Vacuum Valve VV7A:Please seek technical assistance
11	Valve fault - VV7B	Valve fault - Vacuum Valve VV7B:Please seek technical assistance
12	Valve fault - VV10A	Valve fault - Vacuum Valve VV10A:Please seek technical assistance
13	Valve fault - VV10B	Valve fault - Vacuum Valve VV10B:Please seek technical assistance
14	Circuit breaker tripped - BP1	Circuit breaker tripped - Booster Pump 1:Please seek technical assistance
15	Circuit breaker tripped - BP2	Circuit breaker tripped - Booster Pump 2:Please seek technical assistance
16	Circuit breaker tripped - BP3	Circuit breaker tripped - Booster Pump 3:Please seek technical assistance
17	Circuit breaker tripped - DP1	Circuit breaker tripped - Diffusion Pump 1:Please seek technical assistance
18	Circuit breaker tripped - DP2	Circuit breaker tripped - Diffusion Pump 2:Please seek technical assistance
19	Pump Oil Over Temperature - DP1	Pump Oil Over Temperature - Diffusion Pump 1:Please seek technical assistance
20	Pump Oil Over Temperature - DP2	Pump Oil Over Temperature - Diffusion Pump 2:Please seek technical assistance
21	Circuit breaker tripped - CG1	Circuit breaker tripped - CG1:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
22	Circuit breaker tripped - CG2	Circuit breaker tripped - CG2:Please seek technical assistance
23	Circuit breaker tripped - CG3	Circuit breaker tripped - CG3:Please seek technical assistance
24	Cyrogenic fault - CG1	Cyrogenic fault - CG1:Please seek technical assistance
25	Cryogenic fault - CG2	Cryogenic fault - CG2:Please seek technical assistance
26	Cryogenic fault - CG3	Cryogenic fault - CG3:Please seek technical assistance
27	Circuit breaker tripped - RP1	Circuit breaker tripped - Rotary Pump 1:Please seek technical assistance
28	Circuit breaker tripped - RP2	Circuit breaker tripped - Rotary Pump 2:Please seek technical assistance
29	Circuit breaker tripped - RP6	Circuit breaker tripped - Rotary Pump 6:Please seek technical assistance
30	Circuit breaker tripped - RP4	Circuit breaker tripped - Rotary Pump 4:Please seek technical assistance
31	Contactor fail - RP1	Contactor fail - Rotary Pump 1:Please seek technical assistance
32	Contactor fail - RP2	Contactor fail - Rotary Pump 2:Please seek technical assistance
33	Mechanism traverse limit switch fault	Limit switch fault - Mechanism traverse:Please seek technical assistance
34	Over Temperature	Over temperature - Web threader motor:Please seek technical assistance
35	Blower fault	Blower fault - DTR1:Please seek technical assistance
36	Drive fault	Drive fault - DTR1:Please seek technical assistance
37	Low water flow	Low water flow - edge shield:Please seek technical assistance
38	Blower fault	Blower fault - DTR2:Please seek technical assistance
39	Emergency Stop	Emergency Stop has been activated:
40	Fast Stop	Drives have fast stopped:
41	Drive fault	Drive fault - DTR2:Please seek technical assistance
43	Drive fault	Unwind layarm drive is in fault condition:
45	Drive fault	Rewind layarm drive is in fault condition:
47	Blower fault	Blower fault - Metallising drum:Please seek technical assistance
48	Drive fault	Drive fault - Metallising drum:Please seek technical assistance
50	Blower fault	Blower fault - Rewind:Please seek technical assistance
51	Drive fault	Drive fault - Rewind:Please seek technical assistance
52	Over diameter	Rewind roll has exceeded maximum diameter limit:
54	Blower fault	Blower fault - Unwind:Please seek technical assistance
55	Drive fault	Drive fault - Unwind:Please seek technical assistance
56	Over diameter	Unwind roll has exceeded maximum diameter limit:

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
57	Web break	Web break:
58	Counter failed	Counter failed - Drive PLC_E06_0 Panel:Please seek technical assistance
59	Battery failed	Battery failed - Drive PLC_E06_0 Panel:Please seek technical assistance
60	PLC module failed	PLC module failed - Drive PLC_E06_0 Panel:Please seek technical assistance
61	Tensioning failed	Drives have failed to tension on:
62	Rack fault	Rack fault - Drive PLC_E06_0 Panel:Please seek technical assistance
63	Circuit breaker tripped	Circuit breaker tripped - Unwind sidelay:Please seek technical assistance
64	Drive failed	Drive failed - DTR2 roller:Please seek technical assistance
65	Viewing flap fault	View flap has failed to open or is in wrong position:
66	Source low water flow	Source low water flow - Services:Please seek technical assistance
67	Shield low water flow	Shield low water flow - Services:Please seek technical assistance
68	Shutter low water flow	Shutter low water flow - Services:Please seek technical assistance
69	Excessive pressure rise	Excessive pressure rise - Foreline:Please seek technical assistance
70	Excessive pressure rise	Excessive pressure rise - DP1:Please seek technical assistance
71	Excessive pressure rise	Excessive pressure rise - DP2:Please seek technical assistance
72	Drive fault	Drive fault - Unwind sidelay:Please seek technical assistance
73	Circuit breaker tripped	Circuit breaker tripped - Rewind sidelay:Please seek technical assistance
74	Drive fault	Drive fault - Rewind sidelay:Please seek technical assistance
75	Excessive pressure rise	Excessive pressure rise - DP3:Please seek technical assistance
77	Chiller off during run	Chiller is off during the run:
78	DTR1 Chiller off during run	DTR1 Chiller is off during the run:
79	Excessive pressure rise	Excessive pressure rise - VV7A:Please seek technical assistance
80	Excessive pressure rise	Excessive pressure rise - VV7B:Please seek technical assistance
81	Excessive pressure rise	Excessive pressure rise - VV7C:Please seek technical assistance
83	220V circuit breaker tripped	220V circuit breaker tripped - E06_3 Panel:Please seek technical assistance
84	Panel cooling fault	Panel cooling fault - E04 Panel:Please seek technical assistance
85	Panel cooling fault	Panel cooling fault - E06 Panel:Please seek technical assistance
86	Source water failed	Water services have failed for the source:Please seek technical assistance
87	Source contactor fault	Source contactor fault - E04 Panel:Please seek technical assistance
88	Web thickness out of tol-erance	Check web thickness setpoint in recipe
89	Out of wire	Insufficient wire for next run:Replace wire reel(s)
90	Air pressure low	Air pressure low - Main Air Supply:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
91	Panel cooling fault	Panel cooling fault - E03_0 panel:Please seek technical assistance
92	Unwind diameter error	Unwind diameter error:Please seek technical assistance
93	Diffusion pump services fault	Diffusion pump services fault:Please seek technical assistance
94	High Water Pressure	High water pressure - Inlet filter:Please seek technical assistance
95	Water/Air Supply Failed	Water/Air supply has failed for the vacuum pumps:Please seek technical assistance
96	Main Inlet Temperature	Main inlet water feed temperature is below 18C or above 30C:All pumps will turn off in the case of over temperature if this alarm persists for 10 minutes
97	Main Inlet Temperature	Main inlet water feed temperature is below 19C or above 28C:
98	Rewind diameter error	Rewind diameter error:Please seek technical assistance
99	Gas flow failure	Gas flow failure - Precoat plasma treater:Please seek technical assistance
100	Oxygen concentration High	Oxygen concentration high - Precoat plasma treater:Please seek technical assistance
101	Failed to run	Precoat plasma treater has failed to run:
102	Wetting failed	Boats failed to wet down:Switch the wire off and wet down manually
103	Contactor fail	Contactor fail - Rotary Pump 6:Please seek technical assistance
104	Contactor fail	Contactor fail - Rotary Pump 4:Please seek technical assistance
105	Contactor fail / drive fault	Contactor fail / drive fault - Booster Pump 1:Please seek technical assistance
106	Contactor fail / drive fault	Contactor fail / drive fault - Booster Pump 2:Please seek technical assistance
107	Contactor fail / drive fault	Contactor fail / drive fault - Booster Pump 3:Please seek technical assistance
108	Pump fault	Pump fault - Diffusion Pump 1:Please seek technical assistance
109	Pump fault	Pump fault - Diffusion Pump 2:Please seek technical assistance
110	Pump fault	Pump fault - Diffusion Pump 3:Please seek technical assistance
111	Pump fault	Pump fault - Diffusion Pump 4:Please seek technical assistance
112	Circuit breaker tripped	Circuit breaker tripped - Diffusion Pump 3:Please seek technical assistance
113	Circuit breaker tripped	Circuit breaker tripped - Diffusion Pump 4:Please seek technical assistance
114	Pump Oil Over Temperature	Pump Oil Over Temperature - Diffusion Pump 3:Please seek technical assistance
115	Pump Oil Over Temperature	Pump Oil Over Temperature - Diffusion Pump 4:Please seek technical assistance
116	Valve fault - VV6C	Valve fault - VV6C:Please seek technical assistance
117	Valve fault - VV6D	Valve fault - VV6D:Please seek technical assistance
118	Valve fault - VV7D	Valve fault - VV7D:Please seek technical assistance
119	Valve fault - VV7C	Valve fault - VV7C:Please seek technical assistance
120	Circuit breaker tripped - BP4	Circuit breaker tripped - Booster Pump 4:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
121	Pump fault - BP4	Pump fault - Booster Pump 4:Please seek technical assistance
122	Circuit breaker tripped - BP5	Circuit breaker tripped - Booster Pump 5:Please seek technical assistance
123	Pump fault - BP5	Pump fault - Booster Pump 5:Please seek technical assistance
124	Circuit breaker tripped - BP6	Circuit breaker tripped - Booster Pump 6:Please seek technical assistance
125	Pump fault - BP6	Pump fault - Booster Pump 6:Please seek technical assistance
126	Current trip - DP1	Current trip - DP1:Please seek technical assistance
127	Current trip - DP2	Current trip - DP2:Please seek technical assistance
128	Current trip - DP3	Current trip - DP3:Please seek technical assistance
129	IO fault	IO fault - Chiller:Please seek technical assistance
130	IO fault	IO fault - Drives PLC:Please seek technical assistance
131	IO fault	IO fault - Drum Wedge MFC:Please seek technical assistance
132	Spare	Spare - Spare:Please seek technical assistance
133	IO fault	IO fault - E07 Post MFC:Please seek technical assistance
134	IO fault	IO fault - E07 Pre MFC:Please seek technical assistance
135	IO fault	IO fault - E10 Gas Safety:Please seek technical assistance
136	IO fault	IO fault - E03 ET200AL Remote IO:Please seek technical assistance
137	IO fault	IO fault - E03 ET200SP:Please seek technical assistance
138	IO fault	IO fault - E03 Wire Feed:Please seek technical assistance
139	IO fault	IO fault - E04 ET200SP:Please seek technical assistance
140	IO fault	IO fault - E04 Thyristors:Please seek technical assistance
141	IO fault	IO fault - E06 ET200AL Remote IO:Please seek technical assistance
142	Spare	Spare - Spare:Please seek technical assistance
143	IO fault	IO fault - E06 ET200SP:Please seek technical assistance
144	Spare	Spare - Spare:Please seek technical assistance
145	IO fault	IO fault - E07 ET200SP POST:Please seek technical assistance
146	IO fault	IO fault - E07 ET200SP PRE:Please seek technical assistance
147	IO fault	IO fault - CG1:Please seek technical assistance
148	IO fault	IO fault - CG2:Please seek technical assistance
149	IO fault	IO fault - CG3:Please seek technical assistance
150	IO fault	IO fault - CG4:Please seek technical assistance
151	IO fault	IO fault - Rewind Wedge MFC:Please seek technical assistance
152	IO fault	IO fault - T05_5 ET200SP:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
153	IO fault	IO fault - T06_21 MFC:Please seek technical assistance
154	24V distribution fault	24V distribution fault - E03 F3:Please seek technical assistance
155	24V distribution fault	24V distribution fault - E03 F4:Please seek technical assistance
156	24V distribution fault	24V distribution fault - E06_3 F3:Please seek technical assistance
157	24V distribution fault	24V distribution fault - E06_3 F4:Please seek technical assistance
158	Module fault	Module fault - E06_3 ET200SP:Please seek technical assistance
159	IO fault	IO fault - E03-5 ET200SP:Please seek technical assistance
160	IO fault	IO fault - Turbo pump 1:Please seek technical assistance
161	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 1:
162	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 2:
163	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 3:
164	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 4:
165	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 5:
166	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 6:
167	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 7:
168	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 8:
169	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 9:
170	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 10:
171	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 11:
172	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 12:
173	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 13:
174	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 14:
175	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 15:
176	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 16:
177	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 17:

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
178	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 18:
179	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 19:
180	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 20:
181	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 21:
182	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 22:
183	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 23:
184	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 24:
185	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 25:
186	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 26:
187	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 27:
188	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 28:
189	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 29:
190	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 30:
191	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 31:
192	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 32:
193	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 33:
194	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 34:
195	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 35:
196	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 36:
197	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 37:
198	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 38:
199	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 39:

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
200	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 40:
201	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 41:
202	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 42:
203	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 43:
204	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 44:
205	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 45:
206	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 46:
207	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 47:
208	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 48:
209	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 49:
210	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 50:
211	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 51:
212	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 52:
213	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 53:
214	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 54:
215	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 55:
216	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 56:
217	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 57:
218	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 58:
219	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 59:
220	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 60:
221	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 61:

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
222	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 62:
223	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 63:
224	Boat fault or circuit breaker tripped	Boat conduction failed or circuit breaker tripped - Boat 64:
225	Profinet node failure	Profinet node failure - E03-ET200SP Rack:Please seek technical assistance
226	Profinet node failure	Profinet node failure - E04-ET200SP Rack:Please seek technical assistance
227	Profinet node failure	Profinet node failure - E10 ET200SP RACK:Please seek technical assistance
228	Profinet node failure	Profinet node failure - T05 ET200SP RACK:Please seek technical assistance
229	Profinet node failure	Profinet node failure - E03-ET200AL Remote IO:Please seek technical assistance
230	Profinet node failure	Profinet node failure - E06-3-ET200SP Rack:Please seek technical assistance
231	Profinet node failure	Profinet node failure - E07-ET200SP-PRETRTR Rack:Please seek technical assist-ance
232	Profinet node failure	Profinet node failure - E07-ET200SP-POSTTRTR Rack:Please seek technical assist-ance
233	Profinet node failure	Profinet node failure - E06-3-ET200AL Remote IO:Please seek technical assistance
234	Profinet node failure	Profinet node failure - GRE Chiller:Please seek technical assistance
235	Profinet node failure	Profinet node failure - GWK Chiller:Please seek technical assistance
236	Profinet node failure	Profinet node failure - PN-IO-2-1 Rack:Please seek technical assistance
237	Profinet node failure	Profinet node failure - E06-0-ET200SP RACK:Please seek technical assistance
238	Spare	Spare - Spare:Please seek technical assistance
239	Spare	Spare - Spare:Please seek technical assistance
240	Spare	Spare - Spare:Please seek technical assistance
241	High Water Temperature - DP1	High Water Temperature - Diffusion Pump 1:Please seek technical assistance
242	High Water Temperature - DP2	High Water Temperature - Diffusion Pump 2:Please seek technical assistance
243	High Water Temperature - DP3	High Water Temperature - Diffusion Pump 3:Please seek technical assistance
244	High Water Temperature - DP4	High Water Temperature - Diffusion Pump 4:Please seek technical assistance
245	Turbo pump 1 circuit breaker tripped	Turbo pump 1 circuit breaker tripped:Please seek technical assistance
246	Turbo pump 1 fault	Turbo pump 1 fault:Please seek technical assistance
247	Profibus fail	Profibus fail - Unwind layarm:Please seek technical assistance
248	Drive failed	Drive fail - Unwind layarm:Please seek technical assistance
249	Positioning fail	Positioning fail - Unwind layarm:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
250	Diameter measurement fail	Unwind layarm failed to measure diameter:
251	Homing failed	Unwind layarm failed to reach home position:
252	Move to reel failed	Move to reel failed - Unwind layarm:Please seek technical assistance
253	Profibus failed	Profibus failed - Rewind layarm:Please seek technical assistance
254	Drive failed	Drive failed - Rewind layarm:Please seek technical assistance
255	Diameter measure failed	Rewind layarm failed to measure diameter:
256	Homing failed	Rewind layarm failed to reach home position:
257	Move to reel failed	Move to reel failed - Rewind layarm:Please seek technical assistance
258	Profibus failed	Profibus failed - Rewind:Please seek technical assistance
259	Drive failed	Drive failed - Rewind:Please seek technical assistance
260	Profibus failed	Profibus failed - Unwind:Please seek technical assistance
261	Drive failed	Drive failed - Unwind:Please seek technical assistance
262	Profibus failed	Profibus failed - Metallising drum:Please seek technical assistance
263	Drive failed	Drive failed - Metallising drum:Please seek technical assistance
264	Roller slipping	Metallising drum is slipping:
265	Profibus failed	Profibus failed - DTR2 roller:Please seek technical assistance
266	Mechanism traverse drive fault	Mechanism traverse drive fault - E06_3 Panel:Please seek technical assistance
267	Roller slipping	DTR1 is slipping:
268	Roller slipping	DTR2 is slipping:
269	Mechanism traverse circuit breaker tripped	Mechanism traverse circuit breaker tripped - E06_3 Panel:Please seek technical assistance
270	Webpath incorrect around loadcell	Web has been wound incorrectly around the loadcell:
271	Shutter fault	Shutter fault:
272	Wire feeder module failed	Wire feeder module failed:Please seek technical assistance
273	Spare	Spare - Spare:Please seek technical assistance
274	Stack initialisation failed	Stack initialisation failed - AEG Thyristors:Please seek technical assistance
275	Shield drive fault	Shield motor drive fault:Please seek technical assistance
276	Shield limit switch fault	Shield limit switch fault:Please seek technical assistance
277	Shield lowering fault	Shield lowering fault:Please seek technical assistance
278	Spare	Spare
279	Shield raising fault	Shield raising fault:Please seek technical assistance
280	Shield bolt fault	Shield bolt in wrong position:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
281	Shield raise timeout	Shield failed to raise in 10s time limit:Please seek technical assistance
282	Shield lower timeout	Shield failed to lower in 10s time limit:Please seek technical assistance
283	Drive Temperature fault	Temperature fault - Unwind drive:Please seek technical assistance
284	Drive Temperature fault	Temperature fault - Drum drive:Please seek technical assistance
285	Drive Temperature fault	Temperature fault - DTR1 drive:Please seek technical assistance
286	Drive Temperature fault	Temperature fault - Rewind drive:Please seek technical assistance
287	Profibus failed	Profibus failed - DTR1 drive:Please seek technical assistance
288	Drive fault	Drive fault - DTR1 drive E06 Panel:Please seek technical assistance
289	Drive Over Temperature	Overtemperature - DTR2 drive:Please seek technical assistance
290	Drive Over Temperature	Overtemperature - Draw roller drive:Please seek technical assistance
291	Drive fault	Fault - Unwind sidelay:Please seek technical assistance
292	Drive fault	Fault - Rewind sidelay:Please seek technical assistance
293	Sidelay PSU fault	Sidelay PSU fault - E06 Panel:Please seek technical assistance
294	Safety module fault	Safety module fault:3210-A1 - Slot 1 - E03 Panel
295	Air Dilution - Low Gas Flow - RP1	Low air flow through air admit valve - RP1:
296	Air Dilution - Low Gas Flow - RP2	Low air flow through air admit valve - RP2:
297	Air Dilution - Low Gas Flow - RP4	Low air flow through air admit valve - RP4:
298	Air Dilution - Low Gas Flow - RP6	Low air flow through air admit valve - RP6:
299	Safety module fault	Safety module fault:3280-A1 - Slot 19 - E03 Panel
300	Safety module fault	Safety module fault:3285-A1 - Slot 25 - E03 Panel
301	Safety module fault	Safety module fault:3291-A1 - Slot 33 - E03 Panel
302	Safety module fault	Safety module fault:6230-A1 - Slot 2 - E06_0 Panel
303	Safety module fault	Safety module fault:4185-A1 - Slot 2 - E04 Panel
304	Safety module fault	Safety module fault:10110-A1 - Slot 1 - E10 Panel
305	Safety module fault	Safety module fault:10111-A1 - Slot 2 - E10 Panel
306	Safety module fault	Safety module fault:10121-A1 - Slot 3 - E10 Panel
307	Safety module fault	Safety module fault:6410-A1 - Slot 4 - E06_3 Panel
308	Safety module fault	Safety module fault:6465-A1 - Slot 12 - E06_3 Panel
309	Safety module fault	Safety module fault:6470-A1 - Slot 14 - E06_3 Panel
310	Safety module fault	Safety module fault:6475-A1 - Slot 20 - E06_3 Panel
311	Low Water Flow - BP1	Low water flow - Booster Pump 1:Please seek technical assistance

Table 9.2-1 List of Alarms